Abhinav Eluri

Fullstack Developer Python

+918337014233 eluriabhinav@gmail.com https://www.linkedin.com/in/eluriabhinav Bangalore, India

SUMMARY

Full Stack Developer with over 3 years of hands-on experience in web application development and a total of 4.5 years of industry experience. Skilled in **Python(Django), JavaScript, and SQL**, with a focus on delivering high-quality, maintainable code. Familiar with **cloud platforms (AWS, Azure)** and **containerization (Docker)**, with the ability to quickly learn and apply new tools and technologies. Recognized for being a **fast learner, strong debugger**, and continuously adapting to evolving development best practices.

EXPERIENCE

Independant Contractor for a US Client(Individual Contributor)

02/2025 - Present Remote

A technology startup initiative aimed to improve the way of work among teams in an enterprise by achieving alignment through use of products built based on proven methodology

- Developed and maintained **scalable web applications** using **React, TailwindCSS, and shadcn**, ensuring responsive and accessible UI across devices.
- Designed and implemented **database schemas** in **PostgreSQL**, ensuring efficient queries and data integrity for large-scale operations.
- Integrated **real-time features** using **WebSockets** and server-sent events for interactive user experiences.
- Improved system performance by introducing **caching strategies (Redis, Django cache framework)**, lowering API response times significantly.
- **Deployed and managed applications on Render**, leveraging its built-in CI/CD workflows for seamless updates and scaling.

Full Stack Developer

GROON

08/2024 - 02/2025 Hamburg, Germany

Groon's mission is to simplify digital transformation by providing an AI-powered, modular, and cloud-based platform that unifies tools and services into one seamless ecosystem.

- **Developed responsive, dynamic user interfaces** with **React.js, Redux, and Material UI (MUI)**, enhancing usability and customer experience.
- **Built scalable REST APIs** and implemented **real-time communication** features using **Django, DRF, and Django Channels** to support high-performance applications.
- **Integrated third-party marketing services** (Google Ads, Facebook) into a centralized module, enabling automated report generation and KPI tracking.
- **Optimized data workflows** by managing complex datasets (e.g., Google Ads API), improving reporting accuracy and accelerating analytics.

Client Solutions Associate

Flexport GmbH

03/2022 - 03/2023 Hamburg, Germany

Flexport aims to simplify global trade by making logistics transparent, connected, and technology-driven.

- **Streamlined client supply chain operations** by analyzing workflows, resolving issues proactively, and ensuring on-time service delivery.
- **Collaborated with 6+ cross-functional teams** to drive product adoption, enhance client satisfaction, and strengthen long-term business relationships.
- **Developed Python automation scripts** to eliminate repetitive manual tasks and built **PDF data extraction tools**, improving efficiency and accuracy in reporting.
- **Managed financial operations** for 10 key accounts, ensuring accurate invoicing and record maintenance.

Full Stack Developer

Fraunhofer Center for Maritime Logistics

09/2021 - 03/2022 Hamburg, Germany

The institute advances maritime via digital, AI, automation, autonomous systems, and sustainable solutions.

- Increased **vessel monitoring efficiency** by {25%} by developing a real-time ship tracking system using Python, React using sensor information transferred via MQTT.
- Implemented **parallel data pipelines** to process and store sensor streams in **Microsoft SQL Server** while updating a **Flask-based web portal**, ensuring low-latency live monitoring.
- Designed and optimized **database schemas and archival workflows**, including a **Tkinter-based GUI tool** to generate **BACPAC files** and migrate historical data to the cloud, reducing on-premise storage overhead.
- Built an **automated email alerting system** to detect and notify potential collision events, improving operational safety and responsiveness.

Program Analyst

Cognizant Technology Solutions

09/2016 - 03/2018 Bangalore, India

Cognizant Technology Solutions offers global IT services, focusing on digital transformation, cloud, AI, and industry-specific solutions.

- Streamlined global data flow by implementing **Tibco BW** integrations and working closely with the development team to provide feedback and improve process efficiency.
- Partnered with **8 cross-functional teams** and stakeholders to resolve technical issues, reducing downtime and ensuring seamless operations.

EDUCATION

M.Sc. Mechatronics

Technische Universität Hamburg Harburg

10/2018 - 09/2023

B.Tech Mechatronics

SRM Institute Of Science & Technology (Deemed University)

04/2012 - 03/2016

SKILLS

Programming Languages

Python, SQL, JavaScript, Java, Typescript

Frontend

React, Hooks, Redux, HTML5, CSS3, Tailwind

Backend

Python(Django), Python (FastAPI), NodeJS

Containerization & Orchestration:

Docker, Kubernetes

Version Control

Git, GitHub

Databases

Redis, Postgres, SQLite, MongoDB

Microsoft Azure

Machine Learning Designer, Azure AI Vision, Azure Face API, Azure AI language, Language Understanding Service (LUIS)

AI frameworks

Langchain, Langgraph

Vector Databases

Pinecone, Faiss, Chroma

AI Interface Framework

Streamlit, Gradio

ACADEMIC PROJECTS

Master Thesis

Description: This project focuses on data analysis and feature extraction for computer vision, with the objective of building a precise object detection model using the YOLOv8 architecture. The model achieved 99% mean Average Precision (mAP) on the Feldsalat dataset and 71% mAP on the Weeds dataset, demonstrating strong detection accuracy across varied conditions.
Goal: To develop a highly accurate and robust object detection model by applying meticulous data analysis and feature extraction techniques, reinforced through comprehensive error analysis to ensure adaptability in real-world scenarios.
My Role: Led the data-centric aspects of the **MLOps** lifecycle, including data collection, augmentation, preprocessing, and pipeline management. Fine-tuned the YOLOv8 model to improve accuracy and precision, and performed detailed error analysis to enhance overall model robustness.

Lung Disease Classification using ML and Deep Learning techniques

Description: Develop image classification models for lung disease diagnosis using X-ray images, traditional machine learning and Convolutional Neural Networks (CNNs)
Goal: Achieve accurate classification of lung diseases from X-ray images through data preprocessing, feature extraction, model comparison, and real-world applicability
My Role: Responsible for data management, model development, training, evaluation, documentation, and collaboration with healthcare experts for enhanced diagnosis

Improving Crop Yield through Deep Learning-based Plant Health Classification

Description: This project aims to boost crop yield by creating a deep learning model that predicts plant health based on leaf conditions. It involves training various deep learning models and deploying the most effective one for real-time plant health assessment
Goal: Develop an accurate plant health classification model with a focus on leaf images, utilizing deep learning techniques, and deploy it for practical use
My Role: My role includes expertise in TensorFlow, Keras, transfer learning, model evaluation, and model deployment to ensure the successful development and implementation of the plant health classification system

LEGO MAZE SOLVER

Description: Developed a Java-based solution to program a robot for navigating mazes and identifying a target wall in dead ends.
Goal: Designed and tested efficient exploration strategies to ensure the robot could adapt to unknown start positions and maze layouts
My Role: Implemented the maze-solving algorithm, configured the simulator, and optimized performance for accurate navigation.

VOLUNTEERING

HTML and CSS Teacher

Redi school of integration

09/2024 - 12/2024

As a volunteer teaching HTML/CSS classes, I taught the core concepts and shared my expertise with students on how to approach software product development

VOLUNTEERING

Machine Learning and Artificial Intelligence Teacher

Redi school of integration

02/2025 - 06/2025

Taught machine learning fundamentals to 10+ students from diverse backgrounds. Mentored students in practical ML project development

LANGUAGES

German
Advanced

English
Proficient

Hindi
Advanced

PROJECTS

AI Chat Integration

- Developed a comprehensive Al chat platform for interacting with various Al models through a centralized interface
- Real-time communication between React frontend and Django backend using **Server-Side Events (SSE)**
- Integrated diverse Al models, including **OpenAI, Gemini, Claude, Cohere, DeepSeek, and Llama 3.2**
- Supported open-source models via Llama for flexible model selection and individual use
- Implemented global state management and efficient data flow with **Redux**
- Implemented a **user management and authentication system** for secure and personalized interaction

Delivery Management System – FastAPI Backend

- Built a **FastAPI-based**-based Delivery Management REST API using SQL Model, PostgreSQL, and Pydantic with robust validation and relational database design.
- Developed a scalable REST API using FastAPI, SQL Model, and PostgreSQL with strong data validation and relational database design.
- Implemented **OAuth2** authentication with email verification, password resets, and background task processing using **Celery** for email and SMS notifications.

Web-Based Data Analysis and Machine Learning Application

- Developed a user-friendly web app for data analysis using **Pandas** profiling and integrated **Pycaret** for efficient machine learning.
- Streamlined model training, evaluation, and selection with Pycaret and exported the best model in Pickle format for easy deployment.
- Demonstrated expertise in Python, data analysis, **machine learning,** web development, and model deployment.

Document-Based Q&A with Langchain and Streamlit

- Built an interactive PDF question-answering application using **LangChain** and **Streamlit**,enabling users to query documents conversationally.
- Integrated OpenAI and Cohere embeddings with **Pinecone** and **FAISS** vector databases to support scalable and efficient **semantic search**.
- Implemented a context-aware conversational pipeline with memory, improving response accuracy and user experience in document-based chat.

RAG Bot (Retrieval-Augmented Generation) using LangGraph

- Built an intelligent RAG chatbot using LangGraph and LangChain to retrieve and generate accurate answers from structured and unstructured documents.
- Implemented vector embeddings, document chunking, and similarity search using a vector database to improve response accuracy and reduce hallucinations.

FIND ME ONLINE

LinkedIn
eluriabhinav

GitHub
Abhinav-Eluri